

Lab 3 Contributions

- Created separate chat for group members in response to losing the previous one.
- Answered questions from concerned group members about assignment submission.
- Completed and sent all 3 deliverables to group leader via canvas email for assignment submission and repo merge.
- Further assisted other group members

Contribution Journal

Course / Lab: ITAI 1371 — Module 3: Machine Learning Workflow and Types of Learning

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My Individual Contribution

I completed my Module 3 work independently and worked through the machine learning notebook step by step. I opened the notebook in Google Colab, connected the runtime, executed the required cells, and reviewed the resulting outputs.

Data Exploration and Analysis

I examined the Wine dataset and documented the key results: 178 samples, 13 features, three classes, and no missing values. I reviewed the class distribution and feature correlations shown in the visualizations. This helped me understand why data exploration is an important part of the machine learning workflow.

Model Comparison

I reviewed the Logistic Regression and Decision Tree results and compared their accuracy. Logistic Regression achieved 88.9% accuracy, while the Decision Tree achieved 83.3%. I identified Logistic Regression as the better-performing model and explained why I believed it performed better.

Feature Experiment

I completed the feature-selection experiment using alcohol, color intensity, and proline. The selected-feature model achieved 83.3% accuracy compared with the original model's 88.9%. I documented this finding and recognized that additional features may provide useful information for classification.

Assessment and Reflection

I completed the learning-type assessment and received 5/5, or 100%. I also completed the reflection and analysis questions, including the best-performing model, possible reasons for its performance, next steps for improvement, and a real-world application in banking and finance.

Further Investigation

I added questions for further investigation about using all 13 wine features and comparing other algorithms such as Random Forest and Support Vector Machine with Logistic Regression. I also developed answers that explain what could be investigated further without claiming results that were not actually tested.

Documentation and Submission Preparation

I documented my observations in the notebook, reviewed the completed work, downloaded the completed notebook from Google Colab, and uploaded it to my GitHub repository. I also reviewed the assignment requirements and identified the need for the lab PDF, reflection journal, and contribution journal with the required naming convention.

Closing Statement

My contribution consisted of completing the required analysis, model comparison, feature experiment, assessment, reflection, further-investigation questions, and documentation. The process helped me better understand the complete machine learning workflow and how the concepts can apply to practical problems in banking and finance.

Individual Contribution Journal

For this experiment, I independently completed and checked all the cells in the `Module_03_Lab_Exercise.ipynb` notebook and completed the full data analysis process. I personally loaded the Wine dataset and performed exploratory data analysis (EDA), including checking the data shape, class distribution, and correlation matrix of the first six features. I used Pandas, Matplotlib, and Seaborn to complete these tasks and make sure the data was clean and reliable.

During the model building process, I completed the six main steps of machine learning, including data preparation, feature selection, stratified data splitting with an 80/20 ratio, and training logistic regression and decision tree models. During the hands-on session, I also tested different combinations of chemical features by myself. I compared the results and studied how different features affected classification accuracy. I also examined the confusion matrices and classification reports to better understand the results of each model.

For the theoretical assessment, I independently completed the questions about the three main types of machine learning: supervised learning, unsupervised learning, and reinforcement learning. I answered all of the questions correctly. Finally, I organized all the experimental results, adjusted the page layout, and completed the final PDF export and formatting check based on the course requirements.

Lab 03 Contribution Journal

What I Set Out to Do

The task asked me to pick three features from the wine dataset, train a logistic regression model on them, and compare its accuracy against a baseline model that used all features (results['Logistic Regression']).

I chose:

alcohol

color_intensity

proline

These were selected because they're commonly cited as strong discriminators between wine cultivars — proline in particular varies a lot across the three classes in this dataset.

What I Found and Fixed

Bug 1 — tuple indexing instead of list indexing. The starter code had:

```
X_your = df['alcohol', 'color_intensity', 'proline']
```

Passing a comma-separated set of strings inside a single pair of brackets is interpreted by pandas as a tuple, which it tries to look up as one column name — not three separate columns. That raises a `KeyError`. The fix was to wrap the columns in an actual list, and reuse the `your_features` variable I'd already defined instead of retyping the names:

```
X_your = df[your_features]
```

Bug 2 — unscaled features causing convergence issues. `alcohol` sits in the range of roughly 11–15, while `proline` runs into the hundreds or low thousands. Logistic regression's default solver is sensitive to feature scale, and features this mismatched can prevent the optimizer from converging within the default iteration limit — producing a `ConvergenceWarning` and an accuracy score that reflects an unfinished fit rather than the model's real capability.

I addressed this by scaling with `StandardScaler` inside a Pipeline, which also keeps the scaling fit restricted to the training fold (no leakage into the test set):

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.pipeline import make_pipeline
```

```
your_model = make_pipeline(
```

```
StandardScaler(),
```

```
LogisticRegression(random_state=42, max_iter=1000)
```

```
)
```

Fairness of the comparison. I also flagged that comparing a scaled 3-feature model against an unscaled baseline model isn't an apples-to-apples comparison — any accuracy difference could be coming from the preprocessing step rather than the feature choice itself. Both models should use the same preprocessing treatment before their scores are compared.

Result

With the corrected code, the model trains without warnings and produces a stable accuracy score on the held-out test set. (Exact numbers will depend on the actual `random_state=42` split of this dataset run.)

Reflection

Small syntax choices have outsized consequences. The tuple-vs-list bug is an easy typo to make and an easy one to miss when reading code quickly — it's the kind of thing that only becomes obvious once Python raises the `KeyError`.

Preprocessing isn't optional busywork. I initially thought of scaling as a "nice to have," but here it was the difference between a model that converges properly and one that silently gives a misleading score.

Statistical caution matters at small sample sizes. With a test set of only ~36 samples (20% of 178 wine samples), a single misclassified example swings accuracy by about 2.8 percentage points. I learned that a "better" accuracy score from three features vs. the full feature set isn't necessarily meaningful unless the gap is large or backed by cross-validation (e.g., `cross_val_score` averaged over multiple folds) rather than a single train/test split.

Next time, I'd want to run the same 3-feature pipeline through 5-fold cross-validation to get a more trustworthy sense of whether `alcohol`, `color_intensity`, and `proline` are genuinely competitive with the full feature set, or whether the difference I observed was just noise from the particular split.

Contribution Summary